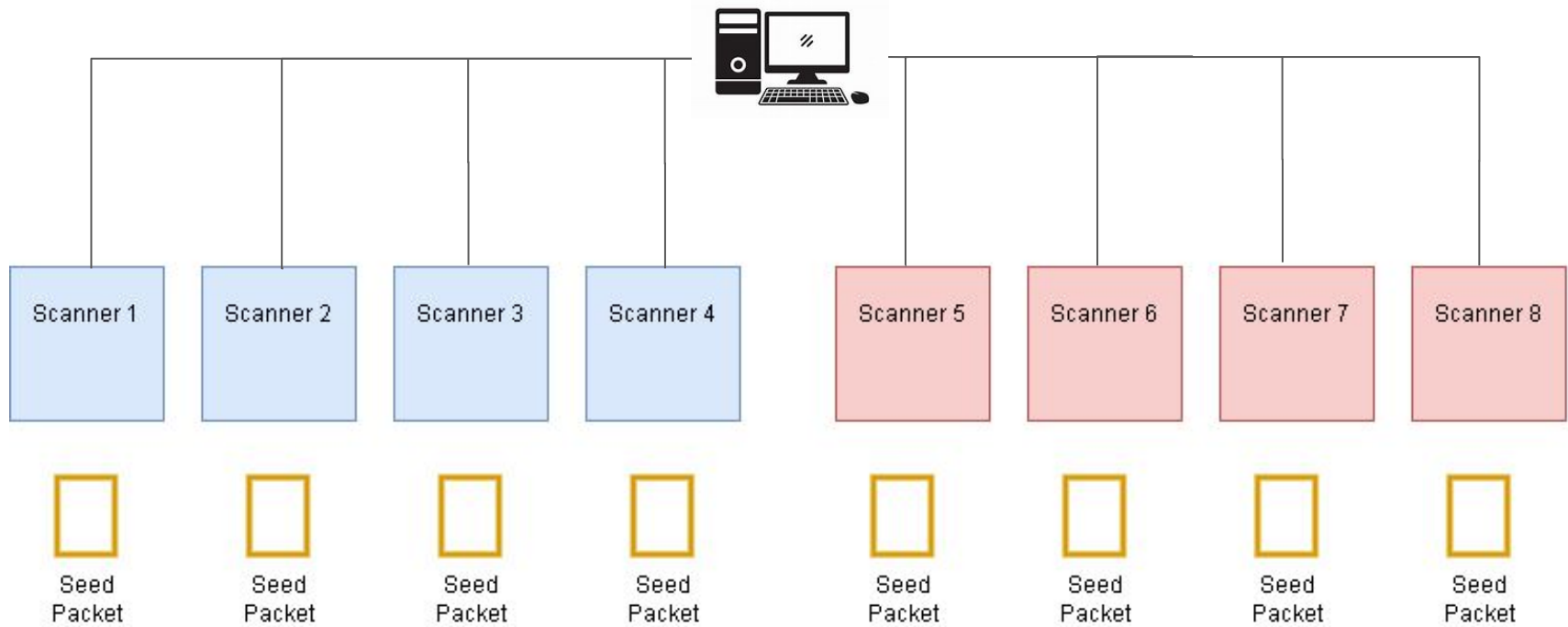


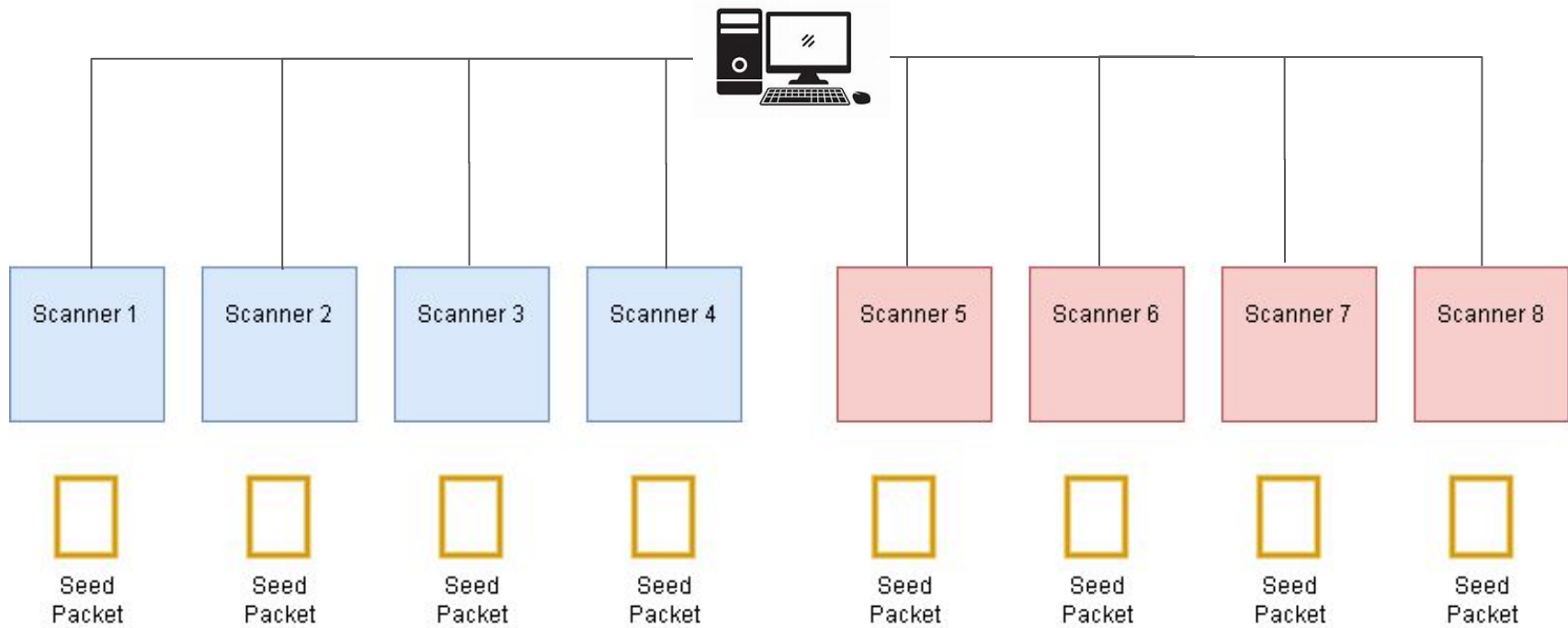
Problem.

You want a fast way to analyze seed yields & health for experimental plots





**Solution: Digitize seed samples with flatbed scanners
then use software to analyze seeds**



Problem: Digitizing data has many bottlenecks that must be addressed

Problem: Digitizing data has many bottlenecks that must be addressed

Detailed scans take up to ~2 minutes to scan, so running 8 scanners sequentially only yields 8 seed scans per 16 minutes, can it be faster?

Yes, run scanners in parallel

How do you track which scans should be activated and exactly which seed packet the scan originated from?

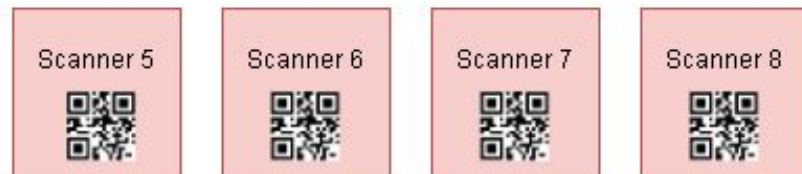
We will use a QR based system to minimize typing:

- QRs on a scanner will be a scanner ID (ie. 'BLUE', 'CRIMSON')
- QRs on the seed packet will correlate to plot information

Batch 1



Batch 2



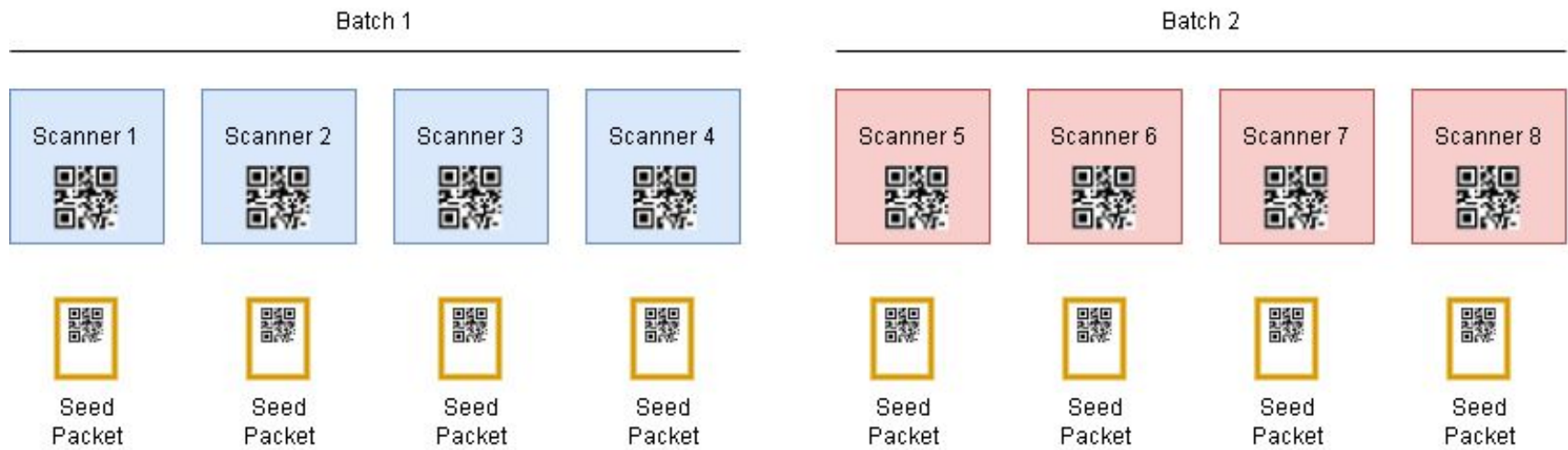
Scanner ID (ie. 'BLUE', 'CRIMSON')

Plot information
from LabelGen



Exp: AMEN
Loc: Lind2024
Variety: MN106-FR
Rep: 1
Pesticide_App: N





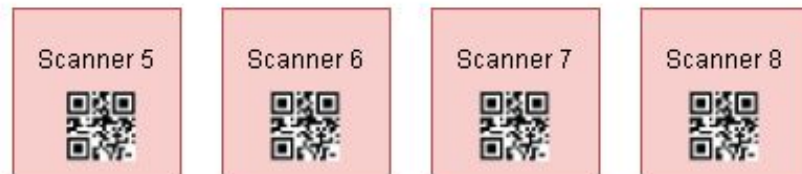
We will split our scanners in two batches, where each one will be operated independently.

This allows for one batch to be loaded/unloaded with seeds while the other batch is scanning (about ~2.5 mins for a batch to complete).

Batch 1

Seed
PacketSeed
PacketSeed
PacketSeed
Packet

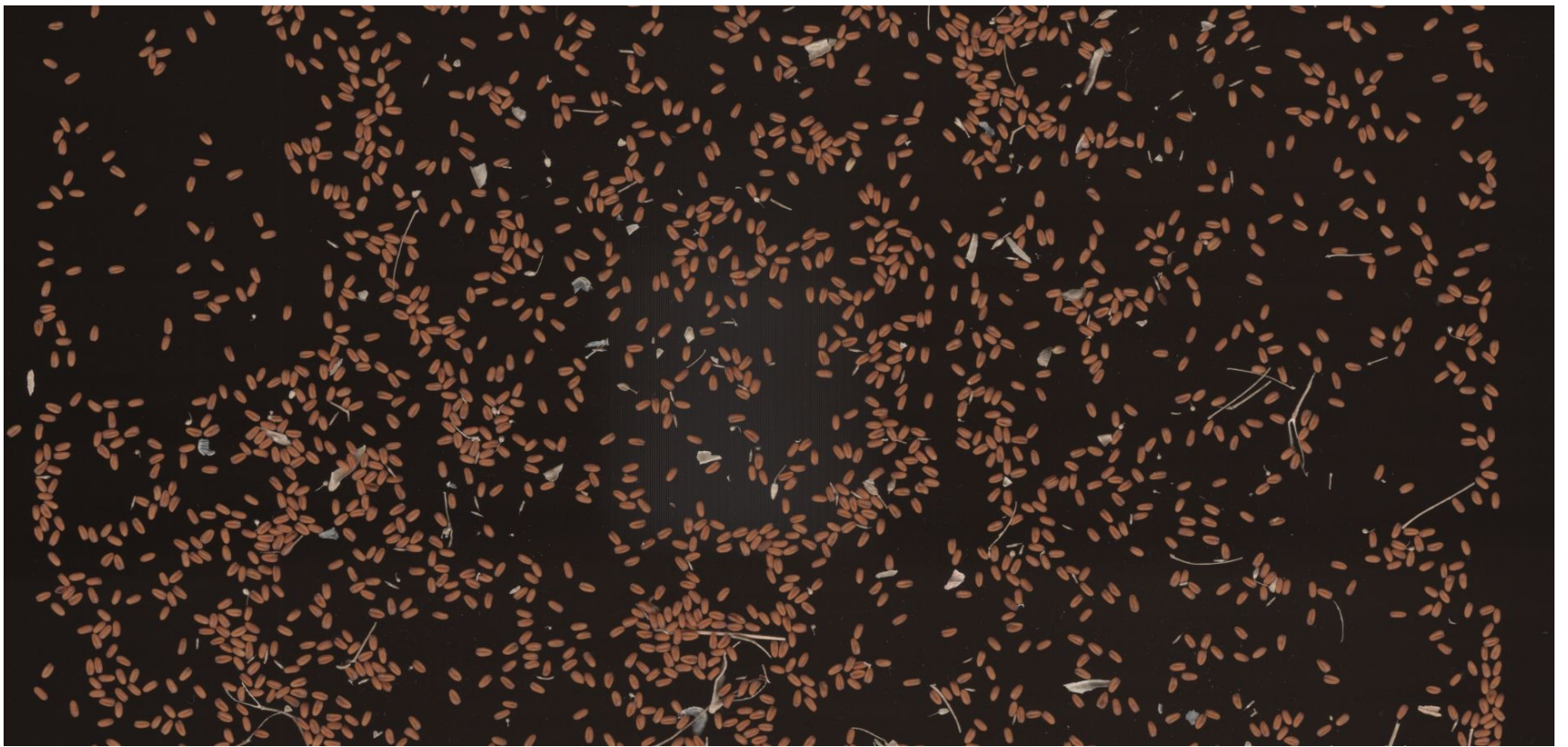
Batch 2

Seed
PacketSeed
PacketSeed
PacketSeed
Packet

Once started, the scanners you wish to run (in a given batch) can be started via COLOR PLOT_INFO pairs

With the QRs, the user simply needs to scan each scanner QR and associated seed packets QR then click enter.





These images can then be given to SeedSizer for quantifiable metrics